Answer $\overrightarrow{OP}$ = ____________ [1]

(ii) Show that O, C and P lie on a straight line.

Answer ____

____ [2]

(c) Find the ratio

(i) area of triangle OCB : area of triangle OCA,

Answer _______ : _______ [1]

(ii) area of triangle OAC : area of triangle OAD**.**

Answer _______ : _______ [1]

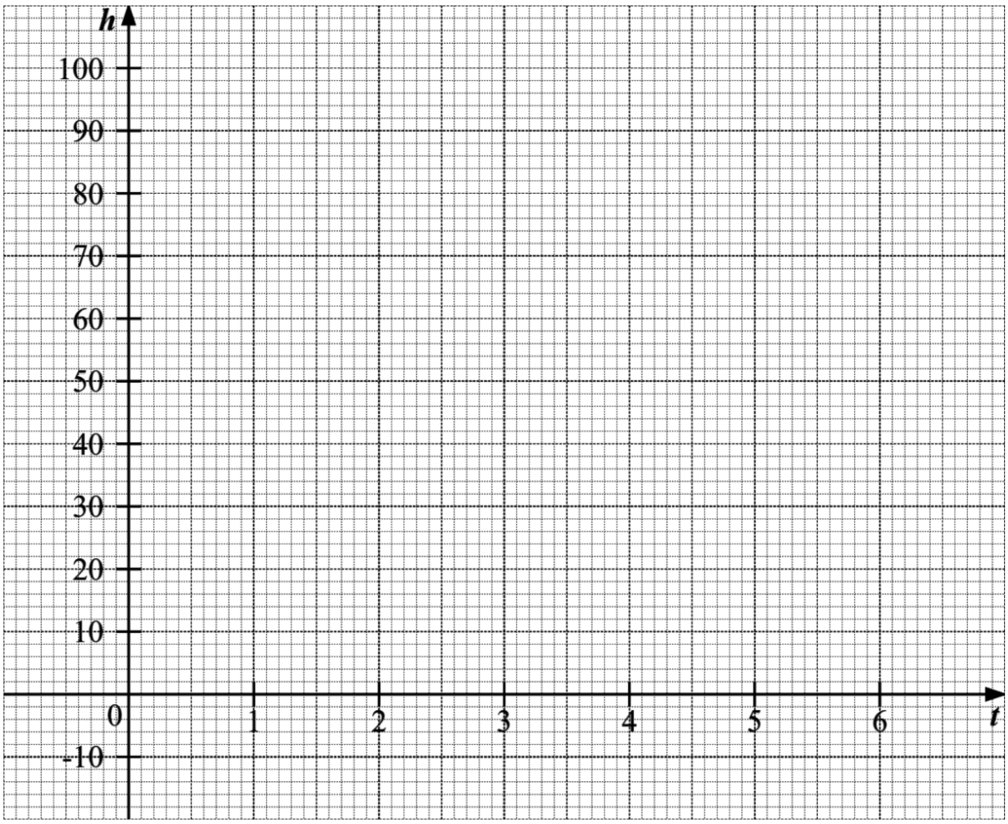
5 A stone is thrown from the top of a cliff next to the sea.
The height, h metres, of the stone above the sea level t seconds after it is released can be modelled by the equation $h = 16t - 5t^2 + 80$.
Some corresponding values of t and h are given in the table below.

t	0	1	2	3	4	5	6
h	80	91	92	83	64	35	p

(a) Calculate the value of p.

Answer $p =$ ______________ [1]

(b) On the grid, draw the graph of $h = 16t - 5t^2 + 80$ for $0 \le t \le 6$.



[3]

(c) Explain how the graph shows that the stone will not reach a height of 100m.

Answer ____

____ [1]

(d) Use the graph to find the length of time that the stone was 84 metres or more above sea level.

Answer ____________________ s [1]

(e) By drawing a tangent on the grid in **part (b)**, find the gradient of the curve at (4, 64). State the units of your answer.

Answer ____________________ [3]